

A New Primary-Source Audit: Two Averaged Selectors Still Do Not Select One Physical Packet

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Abstract

Menon's 2026 Theorems 1.4 and 1.5 sharpen an origin-averaged single-Liouville Fourier bound and a shift-averaged Liouville correlation bound. Their combination does not select the prescribed affine relation and deterministic packet simultaneously. The classification is PRIMARY_SOURCE_AUDIT_L1 and the verdict is SCREENED_NON_DIRECT_ZERO_ELIGIBLE. No program-positive L2 claim or endpoint exponent credit is made.

1 Target contract and source boundary

The target has six simultaneous axes: actual fixed- h_0 packet, source-locked named physical atom, every deterministic prefix, every deterministic scale, a fixed- X power at that atom, and actual active support. The value $h_0 = 2$ is source-backed data only. Repository hashes are integrity locks, not theorem evidence. TPC-193's declared seven-source corpus remains STOP-SCOPED [2].

2 Primary theorem record

Menon's Theorem 1.4 states that, for $X \geq h \geq 10$,

$$\sup_{\alpha \in \mathbb{R}} \frac{1}{X} \int_0^X \left| \frac{1}{h} \sum_{x \leq n \leq x+h} \lambda(n) e(\alpha n) \right| dx \ll \frac{\log \log h}{\log h} + \frac{(\log \log X)^2}{\log X}.$$

Its supremum in α , origin average, $1/X$ and $1/h$ normalizations are all native. Theorem 1.5 states that, for natural $k \geq 2$ and $X \geq H \geq 10$,

$$\frac{1}{H^{k-1}} \sum_{1 \leq h_2, \dots, h_k \leq H} \left| \frac{1}{X} \sum_{1 \leq n \leq X} \lambda(n) \lambda(n+h_2) \cdots \lambda(n+h_k) \right| \ll k \left\{ \frac{\log \log H}{\log H} + \frac{(\log \log X)^2}{\log X} \right\} [1].$$

The outer factor k , both averaging variables, both normalizations, parameter ranges and logarithmic losses are part of the source record.

3 Double-selector lemma

Lemma 1 (Marginals do not select a prescribed cell). *There are arrays $A_{x,h} \in [0, 1]$ for which the average in x at each fixed h and the average in h at each fixed x both tend to zero, while one prescribed entry remains equal to one. Hence separate origin-average and shift-average estimates do not, as a black-box logical operation, control a prescribed origin-shift cell.*

Proof. On an $M \times M$ grid set one prescribed entry to one and all other entries to zero. Every row and column average is at most $1/M$, but the prescribed entry is one. $\square$

This is only a two-axis illustration of the uncontrolled atomic-promotion firewall already locked in TPC-181. It creates no new STOP-SCOPED method cell and is not presented as a new arithmetic obstruction.

4 Six-axis audit

Theorem 1.4 has a phase supremum but one Liouville factor and an averaged origin. Theorem 1.5 has multiple Liouville factors but averaged shifts and no prescribed additive atom on the literal determinant-two packet. Neither theorem supplies the physical coefficient $\mu(d + sz)\mu(u + az)$ together with the production packet, all prefixes, all deterministic scales, a fixed X -power and the full loss ledger. Their juxtaposition does not create a theorem-backed selector.

5 Declared-corpus boundary

TPC193_DECLARED_CANDIDATE_MECHANISM_CORPUS_V1 remains STOP-SCOPED. This paper adds a dated supplemental primary record; it does not rewrite the seven-source V1 corpus or assert global nonexistence. Menon v1 is screened non-direct, with zero eligible literal direct candidates in this supplement.

6 Loss, level and scope ledger

The smallest missing literal theorem is

THEOREM_SELECTING_PRESCRIBED_ORIGIN_SHIFT_RELATION_AND_PACKET.

The next route is NEW_PRIMARY_FIXED_PACKET_SELECTOR_SEARCH. No new method cell is stopped in this paper.

The named-atom exponent credit is 0, while the endpoint requires a strict budget greater than $1/400$; the ledger is unpaid. Block sums are not cumulative prefixes, symbolic packet atoms are not named production atoms, phase L^2 and almost-everywhere statements are not pointwise evaluation, and scoped method failures are not theorem or architecture failures.

7 Machine certificate

The adjacent canonical payload freezes the formula or finite witness, exact repository-source hashes, any explicitly non-hashed external locator record, six target axes, scoped stop, route state and claim firewall. Its recursive exact schema has closed objects, exact array positions and constant leaves. The checker recomputes the finite certificate and executes ten adversarial mutations.

References

- [1] Siddarth Menon. Improved bounds for multiplicative functions in almost all short intervals, 2026.
- [2] Liang Wang. Literal fixed-atom candidate mechanism gate, 2026.